

Ziad Mohamed Shawky

Data Scientist

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SUMMARY

Data Scientist skilled in machine learning, deep learning, and NLP, with hands-on experience across 4 internships and a portfolio of deployed models. Comfortable owning the full workflow: data preprocessing, feature engineering, model development, and deployment using Python, TensorFlow, scikit-learn, and SQL. Built and shipped real NLP and classification systems on FastAPI, Streamlit, and Hugging Face Spaces. Driven by turning messy data into models people can actually use.

SKILLS

Programming: Python, R, SQL, Java

Machine Learning: Scikit-learn, XGBoost, Random Forest, SVM, K-Means, Decision Trees

Deep Learning & NLP: TensorFlow, Keras, PyTorch, CNNs, RNNs, Transformers, spaCy, SBERT, Transfer Learning

Data Analysis & Visualization: Pandas, NumPy, Matplotlib, Seaborn, Power BI, Excel

Databases: MySQL, PostgreSQL, SQLite

Big Data & Streaming: Apache Spark, Hadoop, Apache Kafka, Stream Processing, Distributed Systems (fundamentals)

Deployment & Cloud: FastAPI, Flask, Streamlit, Hugging Face Spaces, AWS

Tools: Git/GitHub, Docker, Google Colab, Jupyter

Core: Feature Engineering, Data Cleaning, Model Evaluation, Hyperparameter Tuning

WORK EXPERIENCE

AI & ML Intern

Jul 2025 – Sep 2025

MAIM Digital Solutions · Alexandria, Egypt

- Developed and optimized 5+ machine learning and deep learning models to address real-world business problems, improving predictive accuracy by ~18% on average.
- Engineered and cleaned 50,000+ structured and unstructured data entries, reducing noise and missing values to improve model reliability.
- Implemented computer vision and NLP solutions using TensorFlow, scikit-learn, Pandas, and NumPy across the full model workflow.
- Built interactive dashboards and visualizations in Power BI and Python to make model outputs clearer for stakeholders.

Machine Learning Intern

May 2025 – Jul 2025

Sprints · Remote

- Built and evaluated 8+ supervised and unsupervised machine learning models, improving prediction accuracy by ~20% on average.
- Performed detailed exploratory data analysis (EDA) and feature engineering to improve model interpretability and reduce training time.
- Cleaned, modeled, and visualized 10,000+ data samples using Python, Pandas, and Matplotlib to ensure high data quality.
- Collaborated with a team of 4 to deliver 3 analytical prototypes addressing real business challenges.

Data Science & AI Intern (DEPI)

Nov 2024 – May 2025

Digital Egypt Pioneers Initiative — New Horizon

- Conducted data preprocessing and feature engineering on 10,000+ real-world samples to improve data quality and model readiness.
- Developed supervised and unsupervised ML models in Python, achieving ~88% average accuracy across projects.
- Used MySQL for database management and analytical querying to speed up data retrieval.
- Created interactive visualizations with Matplotlib and Seaborn to communicate insights to technical and non-technical audiences.

Data Analyst Intern

Jun 2024 – Sep 2024

My Communication · Alexandria, Egypt

- Cleaned and processed 500,000+ data points, improving reporting accuracy and consistency.
- Built 5 interactive Power BI dashboards to support faster, data-driven decisions.
- Automated recurring monthly reports using DAX, cutting manual reporting time significantly.

PROJECTS

HireLens AI-Powered Career Intelligence Platform | [Live Demo](#)2026

- Led a 5-person team and owned the NLP and skill-extraction pipeline (spaCy, SBERT, TF-IDF) for a platform that matches resumes to job descriptions.
- Raised matching precision from 0.71 to 0.89 and F1 from 0.73 to 0.86 via a word-boundary fix in the extraction logic.
- Designed a hybrid scoring engine (SBERT + TF-IDF) with OCR and LLM integration; deployed via FastAPI on Hugging Face Spaces.

BBC News Text Classification | [GitHub](#)Aug 2025

- Built an end-to-end text classification pipeline on the BBC News dataset (2,225 samples, 5 categories).
- Fine-tuned DistilBERT to 97% accuracy, improving macro F1 by 4.5 points over baseline models.
- Deployed a Streamlit app for real-time classification with sub-800 ms response time.

Land Type Classification — Sentinel-2 Imagery | [GitHub](#)May 2025

- Built a deep neural network to classify 6 land types from 10,000+ Sentinel-2 satellite images (team project).
- Achieved 98% accuracy after normalization and noise reduction; deployed on Hugging Face Spaces.

Student Academic Risk Prediction | [GitHub](#)Sep 2025

- Developed an ML pipeline on the UCI Student dataset to classify academic risk levels.
- Reached 84% accuracy with a tuned Random Forest; deployed via FastAPI + Streamlit.

Premier League Data Analysis | [GitHub](#)Nov 2024

- Analyzed 3,000+ matches (2013–2023) to surface trends in team performance and tactical efficiency.
- Queried data via MySQL + SQLAlchemy and visualized xG trends and win–loss ratios with Seaborn and Matplotlib.

EDUCATION

B.Sc. in Data Science

09/2022 – 07/2026

Faculty of Computers and Data Science, Alexandria University · GPA 3.33 / 4.0

- Relevant coursework: Data Mining, Data Science Methodology, Machine Learning, Regression Analysis, Stochastic Processes, Distributed Systems, Stream Processing.

CERTIFICATIONS

- AI & ML Internship — MAIM Digital Solutions (2025)
- Supervised Learning with scikit-learn — DataCamp (2025)
- Machine Learning Internship — Sprints (2025)
- Certified Data Scientist, AI & ML Foundation — DEPI (2025)
- Data Analysis & Big Data — My Communication Academy (2024)

LANGUAGES

English — Professional Proficiency Arabic — Native